

Circuit Design

The Smart Campus Safety System is divided into **three independent monitoring zones**. Each zone uses an **ESP32 DevKit V4** as the main controller to collect sensor data and control local alert devices. Although the sensors vary depending on the monitored environment, all zones use a common output pin configuration for LEDs, buzzer, and relay to simplify software development and future backend integration.

Zone 1 – IoT Lab

Components Used

Component	Purpose
ESP32 DevKit V4	Main Controller
LDR Sensor	Fire Detection
MQ2 Gas Sensor	Gas Leakage Detection
PIR Sensor	Occupancy Detection
Green LED	Safe Status
Yellow LED	Warning Status
Red LED	Critical Status
Buzzer	Audible Alert
Relay Module	Emergency Device Control

Pin Configuration

Component	GPIO
LDR	GPIO34
MQ2	GPIO35
PIR	GPIO32
Green LED	GPIO18
Yellow LED	GPIO19
Red LED	GPIO21
Buzzer	GPIO22
Relay	GPIO23

Zone 2 – Server Room

Components Used

Component	Purpose
ESP32 DevKit V4	Main Controller
LDR Sensor	Fire Detection
HC-SR04	Water/Flood Detection
PIR Sensor	Occupancy Detection
Green LED	Safe Status
Yellow LED	Warning Status

Red LED	Critical Status
Buzzer	Audible Alert
Relay Module	Emergency Device Control

Pin Configuration

Component	GPIO
LDR	GPIO34
HC-SR04 Trigger	GPIO26
HC-SR04 Echo	GPIO25
PIR	GPIO32
Green LED	GPIO18
Yellow LED	GPIO19
Red LED	GPIO21
Buzzer	GPIO22
Relay	GPIO23

Zone 3 – Robotics Lab

Components Used

Component	Purpose
ESP32 DevKit V4	Main Controller
LDR Sensor	Fire Detection
MQ2 Gas Sensor	Gas Leakage Detection
PIR Sensor	Occupancy Detection
Green LED	Safe Status
Yellow LED	Warning Status
Red LED	Critical Status
Buzzer	Audible Alert
Relay Module	Emergency Device Control

Pin Configuration

Component	GPIO
LDR	GPIO34
MQ2	GPIO35
PIR	GPIO32
Green LED	GPIO18
Yellow LED	GPIO19
Red LED	GPIO21
Buzzer	GPIO22
Relay	GPIO23

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